Auteur: Raj Shah
Studierichting: Informatica
Oefeningenreeks 1D nr. 2.1

Opgave: Bereken a, b en c zodanig dat:

$$a\left(z-2\right)+b\left(z^2+3z+1\right)+c\left(2z^2-1\right)=z\left(z+11\right)$$

$$\Leftrightarrow az-2a+bz^2+3bz+b+2cz^2-c=z^2+11z$$

$$\Leftrightarrow \left(b+2c\right)z^2+\left(a+3b\right)z-2a+b-c=z^2+11z$$

$$\Rightarrow \begin{cases} b+2c=1 \\ a+3b=11 \\ -2a+b-c=0 \end{cases}$$
$$\Leftrightarrow \begin{cases} b=1-2c \\ a=11-3b \\ c=-2a+b \end{cases}$$
$$\Leftrightarrow \begin{cases} b=1-2\left(-2a+b\right) \\ a=11-3b \\ c=-2a+b \end{cases}$$
$$\Leftrightarrow \begin{cases} 3b=1+4\left(11-3b\right) \\ a=11-3b \\ c=-2a+b \end{cases}$$
$$\Leftrightarrow \begin{cases} 15b=45 \\ a=11-3b \\ c=-2a+b \end{cases}$$
$$\Leftrightarrow \begin{cases} b=3 \\ a=2 \\ c=-1 \end{cases}$$

References